

Observational insulin sensitivity sits below entered settings in every cohort tested

A replication across 1,679 people in six trials, and what it means for sensitivity derived from device data

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Summary

Sensitivity estimated from what glucose did after insulin was given comes out below the sensitivity people have entered into their pumps. That was established in 2026 on 138 users of open-source closed-loop systems, where the constant anchored to measured sensitivity was 145 against 355 anchored to the profiles people had tuned, a ratio of 0.41. The same work tested whether dosing to the measured value would be safe and found it would not: people dosing well weaker than their measured sensitivity had more hypoglycaemia rather than less, and restricting the estimate to overnight periods left the bias unchanged.

This paper asks whether that result was a property of those 138 people, of the open-source systems they used, or of the way insulin on board was obtained. It was none of them. Across 1,679 people in six trial cohorts, using four different constructions of the estimator and insulin on board reconstructed from delivery records rather than reported by the loop, the ratio of measured to entered sensitivity runs from 0.30 to 0.92 with a median of 0.58. In fifteen estimates it does not once reach 1.0.

What was already known

The earlier analysis reached three conclusions relevant here. The shape of the relationship between sensitivity and total daily dose is derivable from data and scales approximately as one over the square root of daily dose. The level is not: a short-window empirical estimate swings by a third between blocks

and carries almost no memory from one block to the next. And the level that comes out is biased in a specific direction, low, such that dosing to it produces more hypoglycaemia than dosing to a tuned profile.

That last point is the one that matters for interpretation. The gap between measured and entered sensitivity is not evidence that entered settings are wrong. It had already been tested against outcomes and the entered settings won.

What this adds

The cohorts are the Jaeb Center public releases: REPLACE-BG, which ran in 2015 with no algorithm intervening, the Loop observational study, DCLP3, DCLP5 and PEDAP on Control-IQ across ages 2 to adult, and IOBP2 on a bionic pancreas. Ages run from 2 to 82 and daily doses from 7 to 107 U. Insulin on board is reconstructed from the delivery record and an insulin action curve, which is the weaker provenance: the earlier work had the loop's own figure.

Four constructions were used, differing in the decisions that turn out to matter. Whether a correction is counted as a bolus alone or as anything delivered above the programmed basal, which matters because 56% of Loop's correcting insulin and 43% of PEDAP's arrives as raised temporary basal. Whether the denominator is the units given or the insulin action across the window. And whether episodes are required to open with little insulin already on board.

Construction	Cohorts	Ratio, measured to entered
oref archive, loop-recorded insulin on board	1	0.41
Bolus corrections, per unit given	5	0.49 to 0.92
Any route, insulin action	5	0.48 to 0.66
Any route, action, low insulin on board at the start	4	0.30 to 0.65

The spread within a construction is as wide as the spread between them, so the choices above move the answer by about a factor of two and the data cannot adjudicate between them. What survives every choice is the sign.

Why the magnitude cannot be pinned here

Three mechanisms produce a low estimate and this design separates none of them.

Glucose falls on its own from a raised level. Overnight and carbohydrate-free, between 150 and 250 mg/dL, it drops a median 21 to 30 mg/dL over four hours with no correction given, so any estimator that removes that fall is measuring something smaller than a settings file encodes, and any estimator that leaves it in is measuring mean reversion.

Insulin is given when glucose is expected to stay up. Within an identical glucose trace the correlation between insulin committed before a window and insulin delivered during it is 0.054 in the open-loop cohort and 0.447 under a do-it-yourself loop, so a controller keeps responding to something the trace does not carry and its dose stays tied to the outcome.

And the insulin action curve is assumed rather than observed. Reconstructed insulin on board agrees with the pump's own record at a median absolute difference of 0.079 U and a correlation of 0.848 across 188 people in REPLACE-BG, which is good but not exact.

A limit on what a simulation can settle

The obvious way to choose between constructions is to run them against records whose true sensitivity is known. That was done and it does not resolve the question. Simulated open-loop records with unlogged carbohydrate return 51.6 against a true 50 for the bolus-and-units-given construction, but only about 22 qualifying episodes per person, and the simulated controller cannot be made to correct through temporary basal at a realistic rate: at any gain that leaves glucose in a plausible range it never delivers half a unit above schedule in thirty minutes. So the constructions that matter for the closed-loop cohorts cannot be tested against a known answer at all.

Reporting that limitation is the point of this section. An estimator validated on the one case that can be simulated should not be presented as validated for the cases that cannot.

What follows

For anyone setting a pump, nothing changes. The entered constants remain the number to use and the earlier outcome work is the reason.

For anyone deriving sensitivity from device data, which includes every dynamic sensitivity equation in current use, the finding is that the quantity being derived is not the quantity a settings file encodes, and the difference is not small, not confined to one cohort, one algorithm or one age group, and not removed by restricting to overnight. An equation calibrated against observational sensitivity inherits the bias.

For anyone reporting such an estimate, the practical recommendation is to publish the ratio against entered settings alongside the estimate itself. Every construction here looks defensible in isolation and they differ by a factor of two, so the ratio is the diagnostic that makes them comparable.

Method

Analysis is local, in Python, against the Jaeb releases loaded into PostgreSQL 17.9 with TimescaleDB 2.26.1. The corpus holds 132,467,585 CGM readings, 68,995,020 basal records and 5,154,053 bolus records.

Delivery is taken net of the programmed basal throughout, since basal offsets hepatic glucose output rather than lowering glucose. The programmed rate comes from the recorded schedule in Loop and REPLACE-BG, from the 48 half-hourly rates on the case report form in DCLP3, DCLP5 and PEDAP, and from each person's own median delivery at that half hour in IOBP2, which programmes no basal at all.

Entered settings are the per-person median of what was recorded at the time of dosing or on the case report form. They were checked against the pump's own arithmetic: the sensitivity implied by dividing the glucose excess by the recommended correction agrees with the recorded column at a ratio of 1.000 in Loop.

Code is at <https://github.com/tim2000s/dynamic-isf-calculations>. The assembled ratios are produced by `inv009/ratio_evidence.py`, the estimators by `inv009/correction_landing.py` and `inv009/correction_routes.py`, and the simulation check by `inv009/audit_estimators.py`.

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